Menstruation Phase

A Comprehensive Research-Based Clinical Overview

Dysmenorrhea | Fatigue | Iron Loss | Mood | Evidence-Based Management

About This Report

This document synthesises evidence from systematic reviews, Cochrane meta-analyses, WHO data, AAFP guidelines, and NIH/PMC publications on the menstruation phase (Days 1–5 of a 28-day cycle). *Educational use only*.

Compiled May 2026 | Based on PMC, Cochrane, AAFP, WHO & StatPearls sources

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1 What Is the Menstruation Phase?

The menstruation phase marks the first day of the menstrual cycle. When a fertilised egg does not implant, oestradiol and progesterone fall sharply from their luteal peaks to near-baseline levels. This hormonal withdrawal triggers prostaglandin release — primarily PGF2α and PGE2 — from the endometrium, causing uterine smooth-muscle contractions, vasoconstriction, ischaemia, and ultimately the shedding of the uterine lining. In a 28-day cycle this phase spans Days 1–5, though normal bleeding lasts 2–7 days.

Prostaglandins also act systemically on GI smooth muscle (causing nausea and diarrhoea) and on blood vessels (causing headache and dizziness), explaining the broad symptom cluster of the menstruation phase.

Hormonal Profile Across the Menstruation Phase

Day	Oestradiol	Progesterone	Prostaglandins	Primary Effect
Day 1	Very low (↓)	Crashed (↓↓)	Peak (↑↑)	Maximum cramping; bleeding onset; GI effects
Day 2	Very low	Near zero	Very high	Peak pain and heavy flow; systemic symptoms
Day 3	Starting ↑	Near zero	Declining	Bleeding heaviest; some symptom relief
Day 4	Rising ↑	Low	Low–moderate	Pain easing; energy improving
Day 5	Moderate ↑	Low	Low	Bleeding ending; recovery and follicular phase beginning

Key Mechanism: Prostaglandins

Prostaglandin F2α (PGF2α) and E2 (PGE2) are the primary mediators of menstrual pain. Women with dysmenorrhea have significantly higher prostaglandin concentrations in menstrual fluid than pain-free women. NSAIDs — which block prostaglandin synthesis via COX enzyme inhibition — are therefore the most effective first-line treatment. Prostaglandins are highest on Days 1–2 of menses, explaining why pain peaks early.

2 Symptom Prevalence & Epidemiology

Menstrual symptoms are among the most prevalent health complaints globally. The menstruation phase carries a specific, well-studied symptom burden distinct from the luteal phase.

Condition / Symptom	Prevalence	Source
Any menstrual pain (dysmenorrhea)	45–95%	WHO systematic review; multiple meta-analyses
Primary dysmenorrhea (no pelvic pathology)	17–81%	WHO 2006 systematic review (global)
Severe dysmenorrhea	12–14%	WHO systematic review
Dysmenorrhea causing school/work absence	Up to 15%	Primary dysmenorrhea review PMC10309238
Daily activities impacted by period pain	~65.7%	Portugal cross-sectional study
Women seeking medical help for dysmenorrhea	Only 14–28%	India n=1,000 and Portugal studies — grossly under-treated
Fatigue / tiredness during menses	Most common symptom	India dysmenorrhea study (n=1,000)
Nausea during menses	Common (associated)	PMC5585876; StatPearls dysmenorrhea
Diarrhoea during menses	Common (associated)	PMC5585876 — prostaglandin GI effect
Headache during menses	Common (associated)	PMC5585876; India dysmenorrhea study
Sleep disturbance on Days 1–2	~28%	National Sleep Foundation Women and Sleep Poll
Endometriosis (secondary dysmenorrhea)	~10% of reproductive-age women	Global prevalence estimates

Dysmenorrhea is the leading cause of school and work absenteeism among menstruating people worldwide, yet remains undertreated and under-reported.

3 Symptom-by-Symptom Comparison Table

Symptom	Prevalence	Onset / Peak	Severity	Mechanism	Resolution
Lower Abdominal Cramps	45–95%	Day 1–2 (peak Day 1–2)	Moderate–severe; 12–14% severe	PGF2α/PGE2 uterine contractions and ischaemia	Day 3–4
Heavy Bleeding	~1 in 5	Day 2–3 (peak flow)	Varies widely	Endometrial shedding; prostaglandin vasodilation	Day 4–6
Fatigue / Low Energy	Most common overall	Day 1; peaks Day 2–3	Mild–moderate	Blood and iron loss; PG systemic effects; disrupted sleep	Day 4–5
Nausea / Vomiting	Common (associated)	Day 1–2	Mild–moderate	Prostaglandins acting on GI smooth muscle	Day 2–3
Diarrhoea	Common (associated)	Day 1–2 (morning)	Mild	PGF2α stimulates intestinal contractions	Day 1–2
Headache / Migraine	Common (associated)	Day 1; menses migraine pattern	Mild–severe	Oestrogen withdrawal triggers trigeminal sensitisation	Day 2–4
Lower Back Pain	Common (associated)	Day 1–3	Mild–moderate	Uterine contraction radiation; prostaglandin referred pain	Day 3–4
Low Mood / Depression	Moderate	Day 1–2 (transitional)	Mild; resolves quickly	Oestrogen and serotonin bottoming out from late-luteal drop	Day 2–3 as oestrogen rises
Dizziness / Syncope	Rare–uncommon	Day 1–2	Mild–moderate when present	PG vasodilation; low blood pressure; blood loss	Day 2
Breast Tenderness (residual)	Carry-over from luteal	Day 1–2	Mild	Falling progesterone; resolving luteal stimulation	Day 2–3

Sources: PMC5585876; PMC8943241; PMC10309238; StatPearls NBK560834; India study PMC5016343.

4 Severity by Day of Bleeding

Symptom burden follows a predictable trajectory tied to prostaglandin levels, which are highest at bleeding onset and fall over subsequent days.

- **Day 1:** Prostaglandins peak; uterine contractions are strongest; nausea, diarrhoea, headache all peak. Most severe day for most people with dysmenorrhea.
- **Day 2:** Pain and heavy flow persist; systemic symptoms continue. Sleep disturbance most common on nights 1–2 (28% report disruption — National Sleep Foundation).
- **Day 3:** Bleeding often heaviest; prostaglandins decline; pain begins to ease. Iron depletion begins contributing to fatigue.
- **Day 4–5:** Oestrogen starts rising noticeably. Energy and mood typically improve. Most symptoms resolve. GI normalises.

Symptom Intensity by Day of Menses

Symptom	Day 1	Day 2	Day 3	Day 4	Day 5
Cramps	●●●●	●●●●	●●●■	●●■■	●■■■
Fatigue	●●●■	●●●●	●●●■	●●■■	●■■■
Nausea	●●●■	●●■■	●■■■	■■■■	■■■■
Diarrhoea	●●■■	●■■■	■■■■	■■■■	■■■■
Headache	●●●■	●●■■	●■■■	■■■■	■■■■
Back Pain	●●●■	●●■■	●■■■	●■■■	■■■■
Low Mood	●●■■	●■■■	■■■■	■■■■	■■■■
Energy	■■■■	■■■■	●■■■	●●■■	●●●■

● = relative symptom intensity (more dots = higher burden). Energy shown as rising level.

5 Age & BMI Differences

A 2023 systematic review and meta-analysis of 77 studies (PMC9819475) identified eight significant factors associated with primary dysmenorrhea prevalence and severity:

- **Age ≥ 20 years:** OR 1.18 (95% CI: 1.04–1.34). Dysmenorrhea is most prevalent in the second and third decades of life, declining in older women.
- **Low BMI (<18.5 kg/m²):** OR 1.51 (95% CI: 1.01–2.26). Underweight women have significantly higher odds of primary dysmenorrhea.
- **Irregular cycles:** OR 1.28. Cycle irregularity strongly associated with greater menstrual pain.
- **Family history of dysmenorrhea:** OR 3.80 (95% CI: 2.18–6.61). The strongest single risk factor — suggesting significant genetic predisposition.
- **High stress:** OR 1.88. Substantially amplifies pain perception via central sensitisation.
- **Short sleep (<7 hours):** OR 1.19. Late bedtime (>23:00): OR 1.30. Sleep disruption is both a cause and consequence of dysmenorrhea.

Perimenopause Note	Unlike primary dysmenorrhea (which typically improves with age), secondary dysmenorrhea from endometriosis or adenomyosis often worsens or newly presents in the 30s and 40s. Women experiencing worsening period pain in their 30s+ should be evaluated for secondary causes rather than assuming age-related improvement.

6 Evidence-Based Management Strategies

NSAIDs & Pharmacological

NSAIDs are the gold-standard first-line treatment for primary dysmenorrhea, superior to both placebo and acetaminophen in meta-analyses of 35+ RCTs.

- No specific NSAID is superior; ibuprofen, naproxen, and mefenamic acid are all effective options.
- Start 1–2 days before expected menses if periods are predictable — this pre-empts the prostaglandin surge and reduces peak pain.
- Regular scheduled dosing (every 6–8 hours) is more effective than as-needed dosing.
- Approximately 18% of women report little or no relief — these individuals should be evaluated for secondary causes such as endometriosis.
- Hormonal contraceptives (OCP, patch, ring) suppress ovulation and thin the endometrium, reducing prostaglandin production and significantly reducing dysmenorrhea.

Heat Therapy

A 2025 systematic review and meta-analysis (PMC12876241) found that heat therapy achieves comparable analgesic efficacy to NSAIDs with a superior safety profile:

- Apply heat (40°C/104°F) to the lower abdomen for 30–60 minutes. Heat patches or reusable pads are equally effective.
- Mechanism: increases local blood flow, reduces ischaemia, and directly inhibits pain signal transmission via thermoreceptors.
- Particularly useful for those who cannot tolerate NSAIDs (renal issues, GI sensitivity, adolescents).

Other Non-Pharmacological Approaches

- **TENS (Transcutaneous Electrical Nerve Stimulation):** High-frequency TENS reduces pain signals; available as wearable devices; safe for self-use.
- **Exercise:** Gentle movement (yoga, walking) during menstruation improves blood flow and endorphin release. Lower-intensity activities preferred on Days 1–2.
- **Acupuncture:** Small RCTs show benefit; Cochrane evidence is promising but not yet conclusive.
- **Smoking cessation:** Smoking significantly worsens dysmenorrhea; cessation is formally recommended in ACOG-aligned adolescent guidelines.

7 Nutrition & Supplementation During Menstruation

The menstruation phase has specific nutritional demands driven by blood loss, iron depletion, and prostaglandin-mediated inflammation.

Nutrient / Strategy	Target	Evidence	Food Sources
Iron-rich foods	Replace blood-loss iron; prevent fatigue	Strong — iron-deficiency anaemia directly worsens menstrual fatigue	Red meat, spinach, lentils, beans, fortified cereals
Vitamin C (with iron)	Enhance iron absorption	Strong — ascorbic acid triples non-haem iron absorption	Citrus, bell peppers, broccoli, berries
Omega-3 fatty acids	Reduce prostaglandin synthesis	Moderate — fish oil reduces dysmenorrhea severity in RCTs	Oily fish, flaxseed, walnuts, algae oil
Magnesium	Reduce uterine spasm	Moderate — magnesium relaxes smooth muscle	Dark chocolate, pumpkin seeds, nuts, leafy greens
Vitamin D	Anti-inflammatory; reduce pain	Moderate — deficiency associated with worse dysmenorrhea	Oily fish, fortified milk, egg yolks, sunlight
Anti-inflammatory diet	Reduce systemic inflammation	Expert consensus — Mediterranean-style diet reduces prostaglandin burden	Olive oil, fish, vegetables, legumes, whole grains
Limit caffeine	Reduce vasoconstriction and GI irritation	Moderate evidence of worsening cramps and nausea	Limit coffee, tea, cola, energy drinks
Limit alcohol	Reduce inflammation; protect iron levels	Moderate — alcohol worsens prostaglandin-mediated pain	Minimise or avoid during Days 1–3
Ginger	Anti-nausea; anti-inflammatory	Small RCTs show nausea benefit; insufficient for dysmenorrhea per AAFP	Fresh ginger tea, ginger capsules 250 mg

AAFP (2021) considers evidence insufficient to recommend fenugreek, fish oil alone, ginger, valerian, or zinc specifically for dysmenorrhea based on current RCT quality, though omega-3s and magnesium show promising benefit in multiple individual trials.

8 Lifestyle Quick-Reference Tips

NUTRITION	PAIN RELIEF
• Eat iron-rich foods with vitamin C every day (Days 1–5) • Small frequent meals to manage nausea on Days 1–2 • Stay well hydrated — dehydration worsens cramps • Ginger tea for nausea; avoid caffeine and alcohol • Anti-inflammatory foods: berries, oily fish, olive oil, leafy greens	• Start NSAIDs 1–2 days before period if predictable • Heat pad to lower abdomen (40°C) for 30–60 min • TENS device: high-frequency setting for cramp relief • Hot bath or shower for systemic muscle relaxation • Acupressure point SP6 (inside ankle) — small supportive evidence

MOVEMENT	REST & WELLBEING
• Days 1–2: gentle yoga, slow walking, light stretching • Child's pose and supine twists relieve uterine tension • Avoid high-intensity exercise on peak pain days • Days 3–5: gradually increase activity as symptoms ease • Swimming can relieve back and abdominal pain	• Prioritise sleep; heat pad at night reduces cramping disruption • Use a period tracking app to anticipate Day 1 and plan ahead • Apply a warm compress to lower back for referred-pain relief • Communicate your limitations on peak pain days — they are valid • Note any worsening year-on-year — flag to your doctor

9 When to See a Doctor

Primary vs Secondary Dysmenorrhea

Feature	Primary Dysmenorrhea	Secondary Dysmenorrhea
Cause	No identifiable pelvic pathology	Endometriosis, adenomyosis, fibroids, PID
Onset	Within 6–12 months of menarche	Often develops in 20s–40s; or worsens progressively
Pain timing	Day 1–2; improves by Day 3	May persist throughout cycle; worsens around menses
Response to NSAIDs	Usually good (80–82%)	Often partial or poor response
Associated symptoms	GI symptoms, headache, fatigue	Dyspareunia, dyschezia, subfertility
Pelvic exam	Normal	May reveal tenderness, nodularity, or fixed uterus
Imaging	Not routinely required	Transvaginal ultrasound; MRI for deep endometriosis
Treatment	NSAIDs, heat, hormonal contraception	Hormonal suppression, progestins, GnRHa, surgery
Diagnosis delay	N/A	Average 6–8 years to endometriosis diagnosis globally

■ Seek Urgent or Prompt Medical Review If:

- Pain is not controlled by NSAIDs taken at the correct dose and timing
- Period pain worsens year-on-year rather than staying stable or improving
- Pain occurs outside of menstruation (mid-cycle or constant)
- Pain during or after intercourse (dyspareunia) is present
- Bowel symptoms (pain on defecation, rectal bleeding) occur with menses
- Menstrual flow is very heavy (soaking a pad/tampon hourly for 2+ consecutive hours)
- Dizziness, syncope, or very heavy bleeding (may need urgent care)
- Fertility concerns arise — endometriosis causes subfertility and warrants early diagnosis
- Symptoms significantly impair work, school, or daily life despite treatment

10 Research Consensus vs Controversies

Key areas of scientific agreement and active debate in menstruation-phase research.

CONSENSUS	CONTROVERSY
Increased prostaglandins (PGF2α and PGE2) are the central mediators of primary dysmenorrhea, well-established by endometrial biopsy and menstrual fluid studies.	The exact prostaglandin threshold causing symptomatic versus asymptomatic menstruation is undefined. Central sensitisation may be equally important in some women.
NSAIDs are the most effective pharmacological treatment, outperforming placebo and paracetamol in multiple RCTs and meta-analyses.	About 18% of women do not respond adequately to NSAIDs. Whether this reflects inadequate dosing, secondary pathology, or differing pain processing is not fully resolved.
Heat therapy achieves comparable pain relief to NSAIDs with a superior safety profile, per a 2025 meta-analysis (PMC12876241).	Optimal heat temperature, application duration, and device type remain undefined. Most heat-therapy trials are small and short-term.
Family history is the strongest single risk factor for primary dysmenorrhea (OR 3.80), suggesting strong genetic predisposition.	Specific genes or heritable mechanisms driving prostaglandin hyperproduction have not been identified. Twin studies are needed.
Average diagnostic delay for endometriosis is 6–8 years globally — a major healthcare equity failure.	Whether delay is primarily due to patient normalisation of pain, clinician under-recognition, or systemic barriers varies by country and setting.
Low BMI (<18.5) is associated with significantly higher dysmenorrhea risk (OR 1.51).	Causality is unclear: does underweight cause worse dysmenorrhea via hormonal effects, or do women with worse dysmenorrhea eat less due to nausea and become underweight?

11 Key References

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